

WIMBORNE AGCM, AB, Canada

WMO: 719040

Lat: 51.9358N

Lon: 113.5908W

Elev: 3084

StdP: 13.13

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.1	-15.0	-26.4	1.4	-19.7	-21.0	1.9	-14.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	82.0	61.4	78.9	60.3	75.9	59.4	65.6	76.3	63.5	74.6	61.5	72.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	91.6	71.7	59.2	84.1	69.2	57.0	77.6	66.8	32.2	76.0	30.5	74.6	29.0	72.3	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-29.4	88.7	4.1	3.5	-32.3	91.3	-34.7	93.3	-37.0	95.3	-40.0	97.9
			WB	-29.6	70.3	4.0	2.9	-32.5	72.4	-34.8	74.1	-37.1	75.8	-40.0	77.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	17.7	14.5	25.2	37.6	49.9	56.7	61.2	59.9	51.8	39.3	25.4	14.5
	DBStd	20.16	14.66	14.56	13.77	9.18	7.95	5.31	4.49	5.31	8.24	9.17	12.98	13.99
	HDD50	5519	1002	994	769	380	102	10	0	2	74	347	739	1100
	HDD65	9915	1467	1414	1234	823	471	254	134	171	398	796	1188	1565
	CDD50	1115	0	0	0	7	98	209	346	309	128	16	2	0
	CDD65	35	0	0	0	0	1	3	15	13	3	0	0	0
	CDH74	1074	0	0	0	5	91	111	326	357	178	6	0	0
	CDH80	194	0	0	0	0	7	11	60	78	38	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	20.3	0.7	0.6	1.0	1.5	2.7	3.9	2.9	2.5	1.8	1.0	0.9	0.8
(15) Precipitation	PrecMax	26.7	1.5	1.3	2.1	3.4	4.3	10.0	6.3	4.5	4.3	2.0	2.4	1.7
	PrecMin	12.9	0.2	0.1	0.3	0.6	0.9	1.7	1.1	0.5	0.3	0.3	0.2	0.2
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.0	1.5	1.2	0.9	0.9	0.4	0.5	0.4
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.0	1.5	1.2	0.9	0.9	0.4	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	46.3	57.6	71.7	81.0	81.0	85.6	86.7	84.7	72.8	61.4	46.5
		MCWB	37.3	38.2	43.3	49.8	55.3	60.4	64.8	62.0	58.1	51.4	46.9	37.5
	2%	DB	43.7	41.4	50.9	65.2	76.7	77.3	81.1	81.4	79.2	64.7	52.6	40.9
		MCWB	36.4	35.0	40.1	47.3	53.5	59.3	64.7	62.1	56.8	48.4	41.8	35.0
	5%	DB	41.0	37.6	46.0	58.9	71.7	73.6	77.5	78.0	74.5	59.7	46.3	37.8
		MCWB	34.9	32.4	37.5	43.6	51.6	57.3	63.3	61.4	55.7	46.4	37.9	32.2
	10%	DB	37.9	34.3	41.7	53.4	67.0	70.2	74.3	74.2	68.7	54.5	41.7	33.8
		MCWB	32.7	30.2	35.1	40.7	49.9	55.3	61.6	60.3	53.3	43.5	35.3	29.5
	0.4%	WB	38.5	38.3	45.4	50.8	56.6	64.7	69.7	67.4	60.4	53.4	46.8	37.9
		MCDB	45.0	45.2	55.9	70.5	76.7	76.9	79.9	77.8	78.1	70.1	59.6	43.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.8	35.3	41.1	47.4	54.5	61.4	66.7	65.0	58.4	50.0	42.1	35.1
		MCDB	42.6	40.6	49.7	64.3	72.6	73.3	76.9	76.2	75.3	62.6	50.9	40.6
	5%	WB	35.1	32.8	38.0	44.4	52.7	59.0	64.4	63.0	56.2	47.1	38.8	32.4
		MCDB	40.7	37.0	45.6	56.9	68.9	69.6	75.4	74.8	71.3	58.1	45.4	37.2
	10%	WB	32.6	30.5	35.4	41.5	50.9	56.9	62.5	60.9	54.1	44.5	35.7	29.5
		MCDB	37.8	34.1	41.1	51.9	65.1	67.8	72.5	72.4	67.8	53.3	40.5	33.9
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	18.2	18.5	21.2	23.9	21.9	22.8	24.4	24.7	21.8	18.1	17.0
		MCDBR	17.8	20.9	23.4	33.0	33.0	29.1	29.7	32.7	36.2	30.0	24.1	20.1
		MCWBR	12.6	15.5	14.6	17.5	14.6	14.0	16.4	16.6	16.8	16.5	15.4	15.0
	5% WB	MCDBR	18.0	20.2	22.6	31.0	28.8	25.3	28.0	29.0	32.9	28.7	23.5	19.7
		MCWBR	13.3	15.3	14.9	16.8	13.4	13.5	16.4	16.1	15.7	16.0	15.6	15.2
(40) Clear-Sky Solar Irradiance	taub		0.259	0.270	0.284	0.325	0.337	0.341	0.342	0.356	0.309	0.280	0.251	0.241
	taud		2.342	2.404	2.449	2.366	2.389	2.408	2.419	2.372	2.471	2.536	2.480	2.349
	Ebn at Noon		243	276	293	290	290	289	287	276	280	270	250	232
	Edh at Noon		19	24	28	35	36	35	34	34	27	21	16	16
	RadAvg		345	671	1110	1496	1812	1927	2045	1665	1166	701	367	264
(45) All-Sky Solar Radiation	RadStd		28	55	87	127	119	121	106	116	124	81	32	25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page